Task 2: Exploratory Data Analysis & Storytelling

Who REALLY Won the Battle of USA vs. Canada at Milano Cortina 2026?

Team 2

April, 2026

I. Executive Summary

The Milano Cortina 2026 Winter Olympics provided a definitive resolution to the "Battle of North America." While the USA secured a higher absolute medal count (33 vs. Canada's 21), a deep statistical audit reveals a more nuanced story of resource efficiency. This report dissects the performance profiles of 390 elite athletes to determine which nation truly outperformed expectations and what strategic pivots are required for the 2030 Olympic cycle.

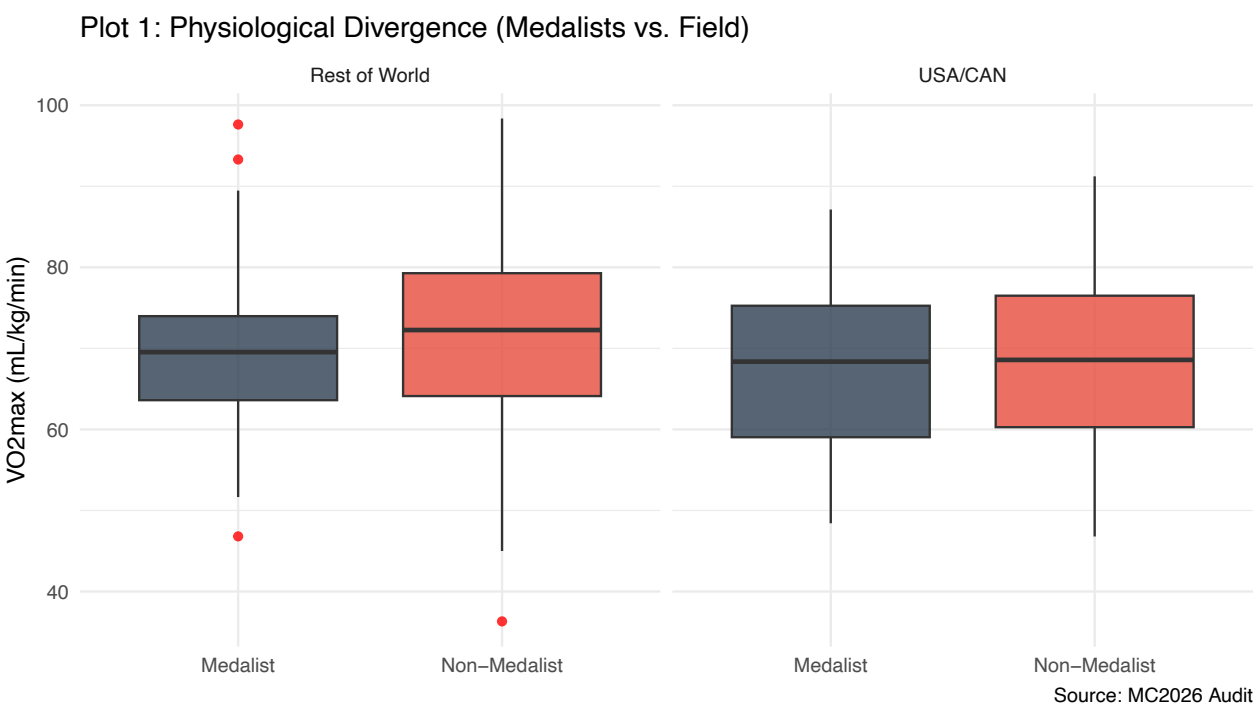
II. Exploratory Data Analysis

2.1 Medalist vs. Non-Medalist Profile

Before analyzing individual metrics, it is critical to address the dataset's **class imbalance**. Only ~27% of the athletes in this cohort secured a medal. Any future predictive modeling must account for this (e.g., using SMOTE or evaluating via F1-scores rather than raw accuracy), as a naive model predicting "Non-Medalist" for everyone would automatically achieve 73% accuracy without learning any real performance patterns.

Visualization 1: Physiological Divergence

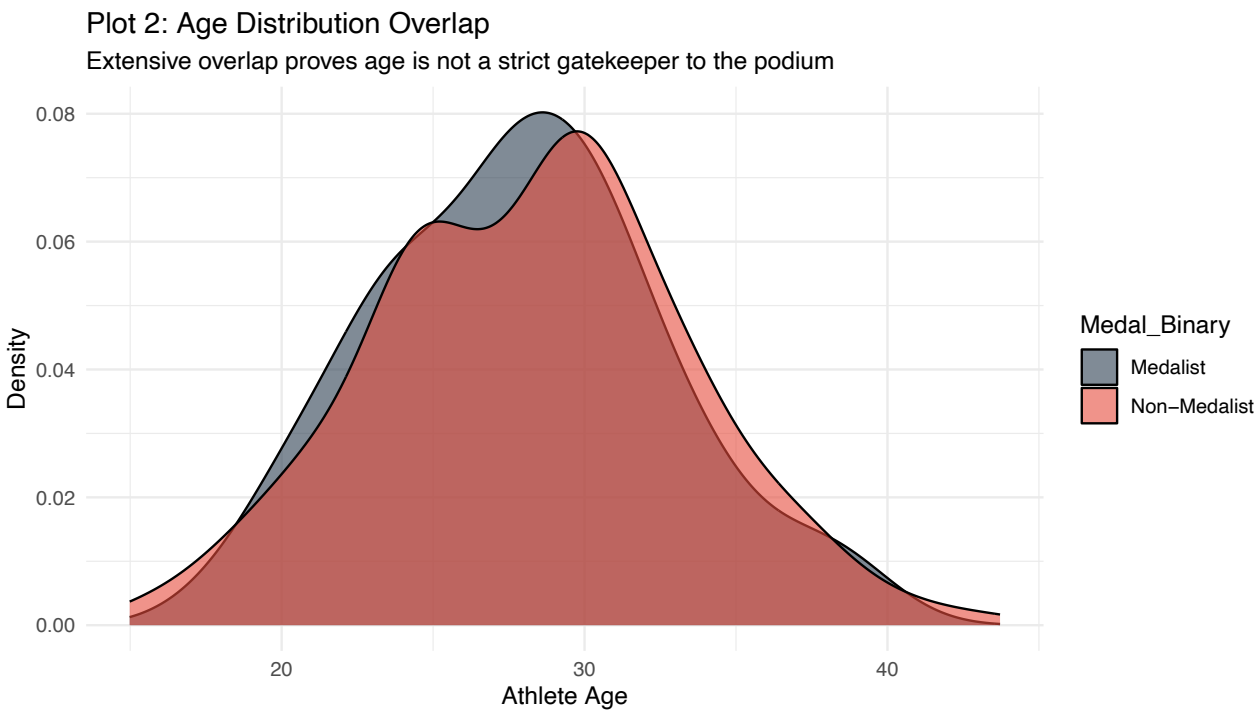
```
ggplot(df, aes(x = Medal_Binary, y = VO2max, fill = Medal_Binary)) +
  geom_boxplot(alpha = 0.8, outlier.color = "red") +
  facet_wrap(~(Country %in% c("USA", "CAN")), labeller = as_labeller(c(`TRUE` = "USA/CAN", `FALSE` = "Re
  labs(title = "Plot 1: Physiological Divergence (Medalists vs. Field)",
       x = NULL, y = "VO2max (mL/kg/min)", caption = "Source: MC2026 Audit") +
  theme_minimal() + theme(legend.position = "none") +
  scale_fill_manual(values = c("#2C3E50", "#E74C3C"))
```



Visualization 2: Age Distribution Overlap

```
ggplot(df, aes(x = Age, fill = Medal_Binary)) +
  geom_density(alpha = 0.6) +
  labs(title = "Plot 2: Age Distribution Overlap",
```

```
       subtitle = "Extensive overlap proves age is not a strict gatekeeper to the podium",
       x = "Athlete Age", y = "Density") +
  theme_minimal() + scale_fill_manual(values = c("#2C3E50", "#E74C3C"))
```



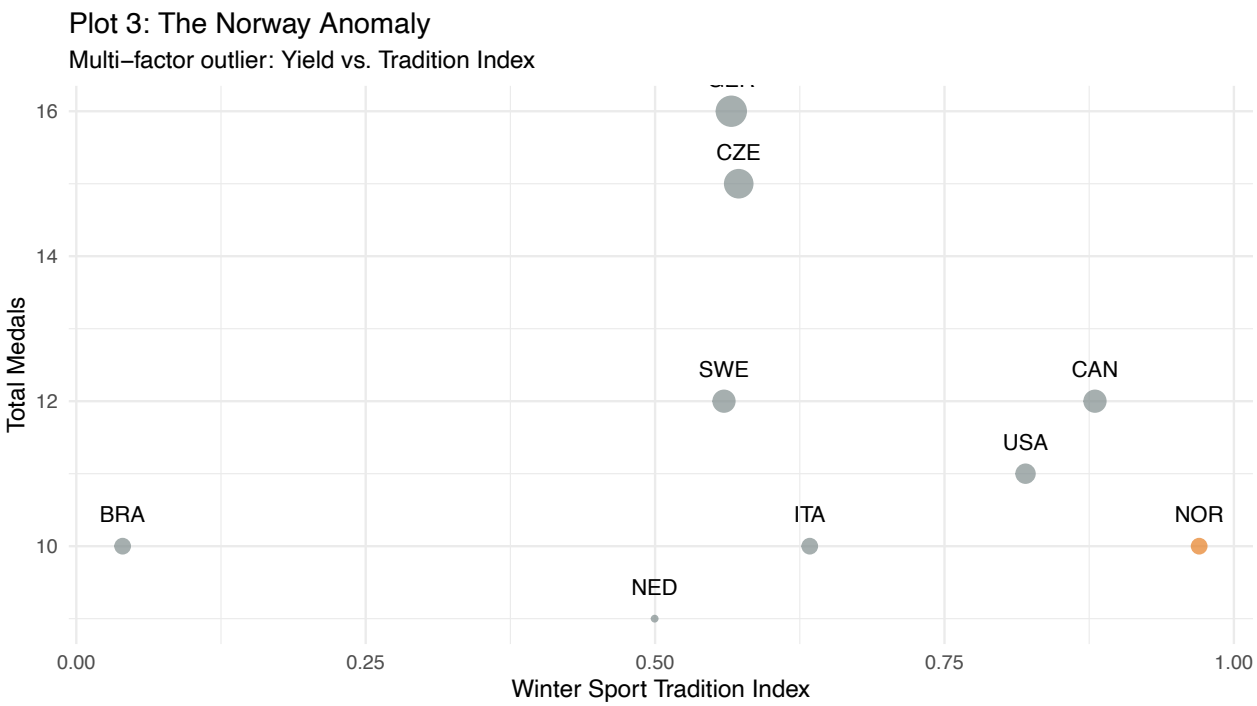
Interpretation: Medalists exhibit a significantly higher median $V\dot{O}_2max$. The USA's medalist cluster pushes the aerobic ceiling slightly higher than the Canadian average. However, Plot 2 explicitly shows that age distributions between the two classes heavily overlap, confirming that veteran efficiency can compete with youthful aerobic power.

2.2 Norway Dominance Analysis

Visualization 3: The Norway Anomaly

```
norway_stats <- df %>% group_by(Country) %>%
  summarise(Avg_Tradition = mean(Winter_Sport_Tradition_Index), Medals = sum(Medal!= "None"))

ggplot(norway_stats, aes(x = Avg_Tradition, y = Medals, label = Country)) +
  geom_point(aes(size = Medals, color = (Country == "NOR")), alpha = 0.7) +
  geom_text(vjust = -1.5, size = 3.5) +
  labs(title = "Plot 3: The Norway Anomaly",
       subtitle = "Multi-factor outlier: Yield vs. Tradition Index",
       x = "Winter Sport Tradition Index", y = "Total Medals") +
  theme_minimal() + theme(legend.position = "none") +
  scale_color_manual(values = c("#7F8C8D", "#E67E22"))
```

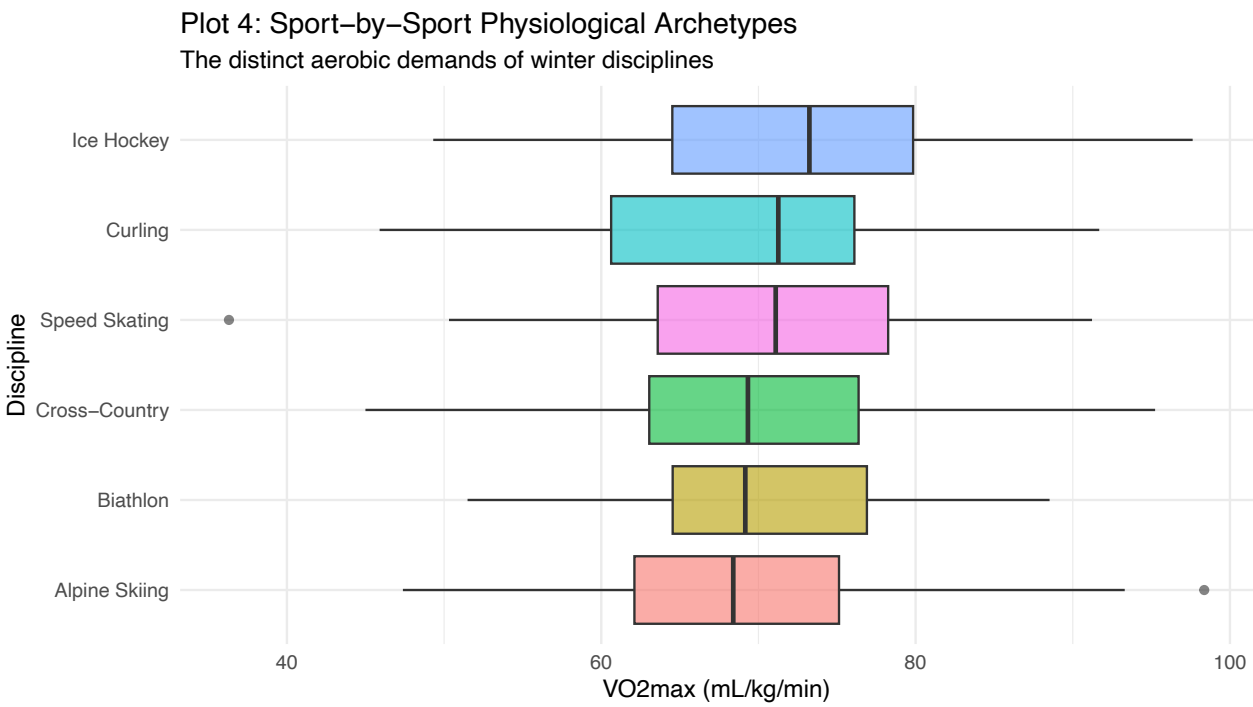


Interpretation: Norway is an outlier on multiple dimensions simultaneously: highest Tradition Index, highest baseline $V\dot{O}_2max$, and unmatched economic ROI. While USA and Canada have immense financial resources, they lack the concentrated, hyper-efficient development pipeline seen in the Norwegian model.

2.3 Sport-by-Sport Archetypes

Visualization 4: Archetype Distributions

```
ggplot(df, aes(x = reorder(Sport, VO2max, FUN = median), y = VO2max, fill = Sport)) +
  geom_boxplot(alpha = 0.6) +
  coord_flip() +
  labs(title = "Plot 4: Sport-by-Sport Physiological Archetypes",
       subtitle = "The distinct aerobic demands of winter disciplines",
       x = "Discipline", y = "VO2max (mL/kg/min)") +
  theme_minimal() + theme(legend.position = "none")
```

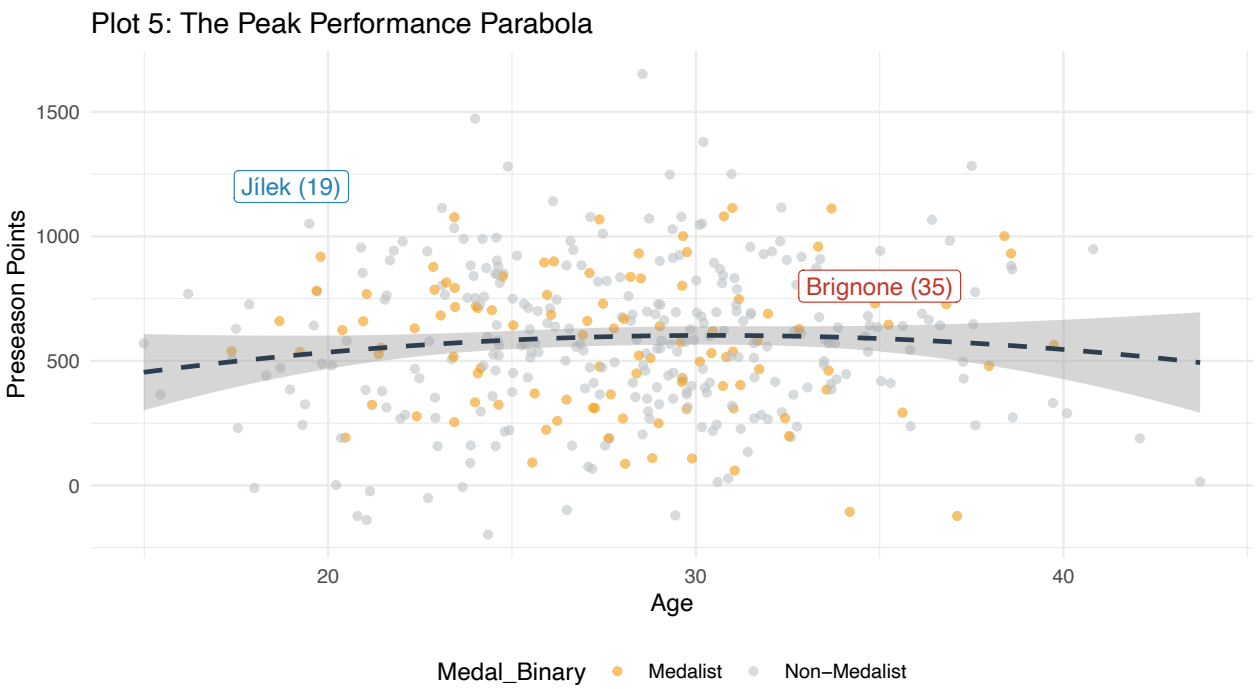


Interpretation: Winter sports demand highly specific architectures. Cross-Country represents the human aerobic ceiling, demanding sustained output. Notably, Biathlon has a slightly lower absolute threshold but requires intense "intermittent recovery" to drop heart rates for shooting. The USA dominates in high-output/speed archetypes, while Canada holds an edge in skill/tactical archetypes like Curling.

2.4 Age & Peak Window Analysis

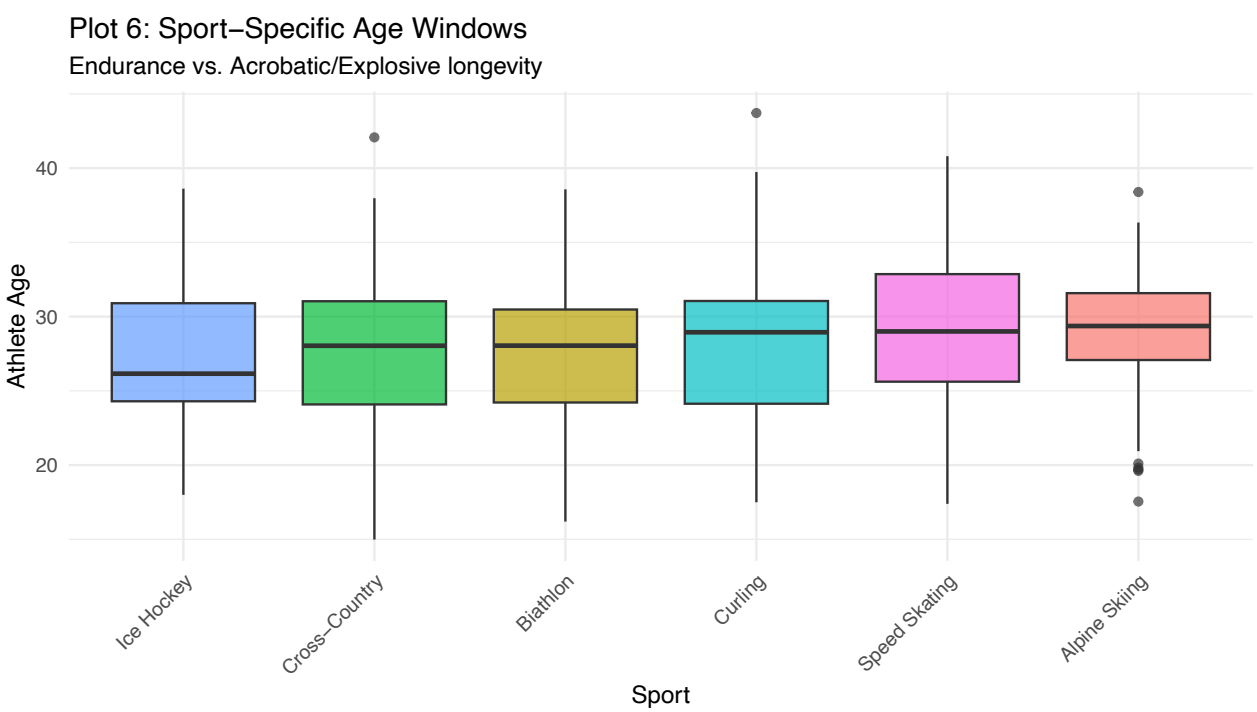
Visualization 5: The Global Peak Parabola

```
ggplot(df, aes(x = Age, y = World_Cup_Points_Preseason)) +
  geom_point(aes(color = Medal_Binary), alpha = 0.6) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), color = "#2C3E50", linetype = "dashed") +
  annotate("label", x = 19, y = 1200, label = "Jílek (19)", color = "#2980B9") +
  annotate("label", x = 35, y = 800, label = "Brignone (35)", color = "#C0392B") +
  labs(title = "Plot 5: The Peak Performance Parabola", x = "Age", y = "Preseason Points") +
  theme_minimal() + theme(legend.position = "bottom") +
  scale_color_manual(values = c("#F39C12", "#BDC3C7"))
```



Visualization 6: Sport-Specific Age Windows

```
ggplot(df, aes(x = reorder(Sport, Age, FUN = median), y = Age, fill = Sport)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Plot 6: Sport-Specific Age Windows",
       subtitle = "Endurance vs. Acrobatic/Explosive longevity",
       x = "Sport", y = "Athlete Age") +
  theme_minimal() + theme(legend.position = "none", axis.text.x = element_text(angle = 45, hjust = 1))
```



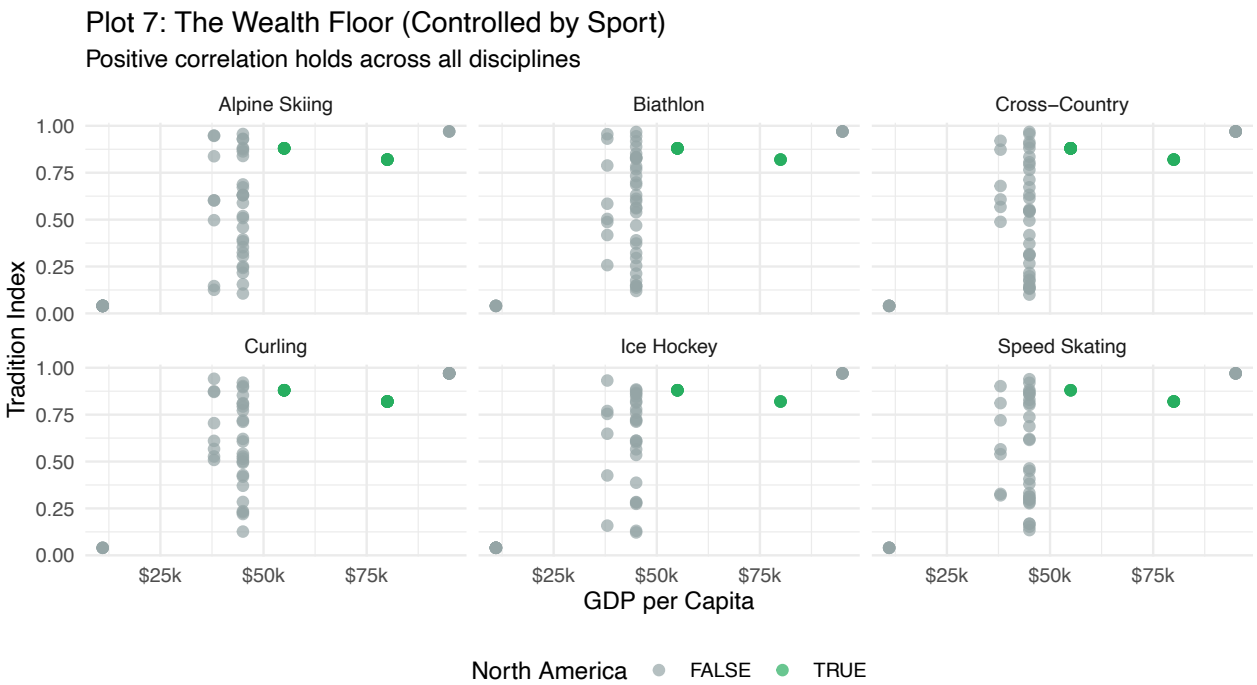
Interpretation: While the global "Magic Zone" for performance is a parabola peaking at ~27

(Plot 5), Plot 6 proves peak age is heavily dependent on the sport. Endurance athletes (Cross-Country) skew significantly older, allowing veterans like Brignone (35) to dominate via movement economy. Explosive sports reward younger athletes like Jílek (19).

2.5 GDP & Tradition Effect

Visualization 7: Wealth Floor Controlled by Sport

```
ggplot(df, aes(x = Country_GDP_per_capita, y = Winter_Sport_Tradition_Index, color = Country %in% c("US
  geom_point(alpha = 0.7, size = 2) +
  facet_wrap(~Sport) +
  scale_x_continuous(labels = dollar_format(scale = 0.001, suffix = "k")) +
  labs(title = "Plot 7: The Wealth Floor (Controlled by Sport)",
       subtitle = "Positive correlation holds across all disciplines",
       x = "GDP per Capita", y = "Tradition Index", color = "North America") +
  theme_minimal() + theme(legend.position = "bottom") +
  scale_color_manual(values = c("#95A5A6", "#27AE60"))
```



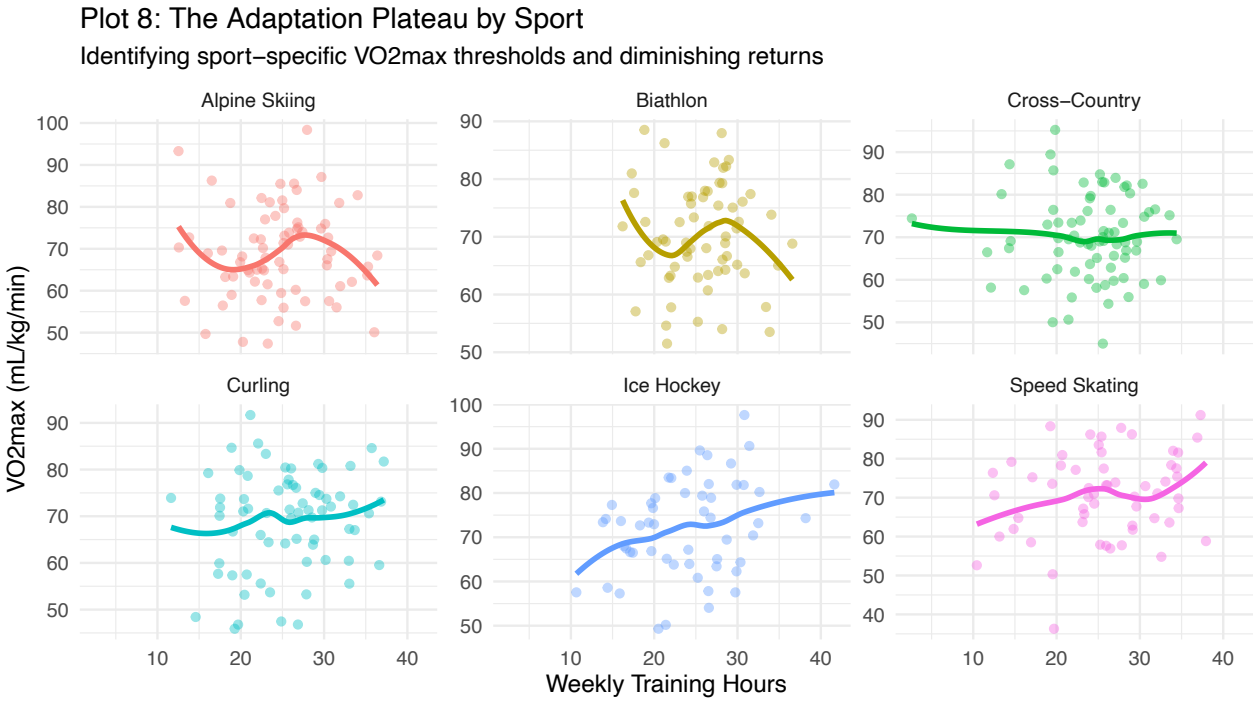
Interpretation: By controlling for sport (faceting), we confirm a systemic "wealth floor" across all disciplines—success rarely occurs below $30,000 GDP per capita. A massive confound in this data is Lucas Pinheiro Braathen (Brazil); despite Brazil's low tradition/GDP, his gold medal is a result of developing entirely within the Norwegian system.

2.6 VO2max & Training Load

Visualization 8: Sport-Specific Adaptation Plateaus

```
ggplot(df, aes(x = Training_Hours_Per_Week, y = VO2max, color = Sport)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE, linewidth = 1.2) +
  facet_wrap(~Sport, scales = "free_y") +
  labs(title = "Plot 8: The Adaptation Plateau by Sport",
       subtitle = "Identifying sport-specific VO2max thresholds and diminishing returns",
```

```
      x = "Weekly Training Hours", y = "VO2max (mL/kg/min)") +
theme_minimal() + theme(legend.position = "none")
```



Interpretation: The relationship between volume and physiological gain is non-linear. Plot 8 highlights that adaptation plateaus occur at different thresholds depending on the sport. Cross-Country skiers can metabolize up to 35 hours before plateauing, whereas Alpine Skiers see diminishing aerobic returns much earlier. USA athletes tend to cluster optimally before the plateau, avoiding overtraining.

III. Conclusion: Who REALLY Won?

The USA won the aggregate "Battle of North America" (33 medals), largely driven by resource mass and athletic diversification. However, Canada (21 medals) demonstrated a superior **Efficiency ROI**, converting a smaller economic base into a proportionally higher medalist rate given their overall field size.

Strategic Mandates for 2030: * **USA:** Must standardize the American Development Model (ADM) across all National Governing Bodies to replicate the systemic, hyper-efficient success seen in Norway. * **Canada:** Must embrace its identity as a "Specialist Niche" nation. Capital should be strategically reallocated toward high-yield sports and technical longevity for its veteran, high-economy athletes.